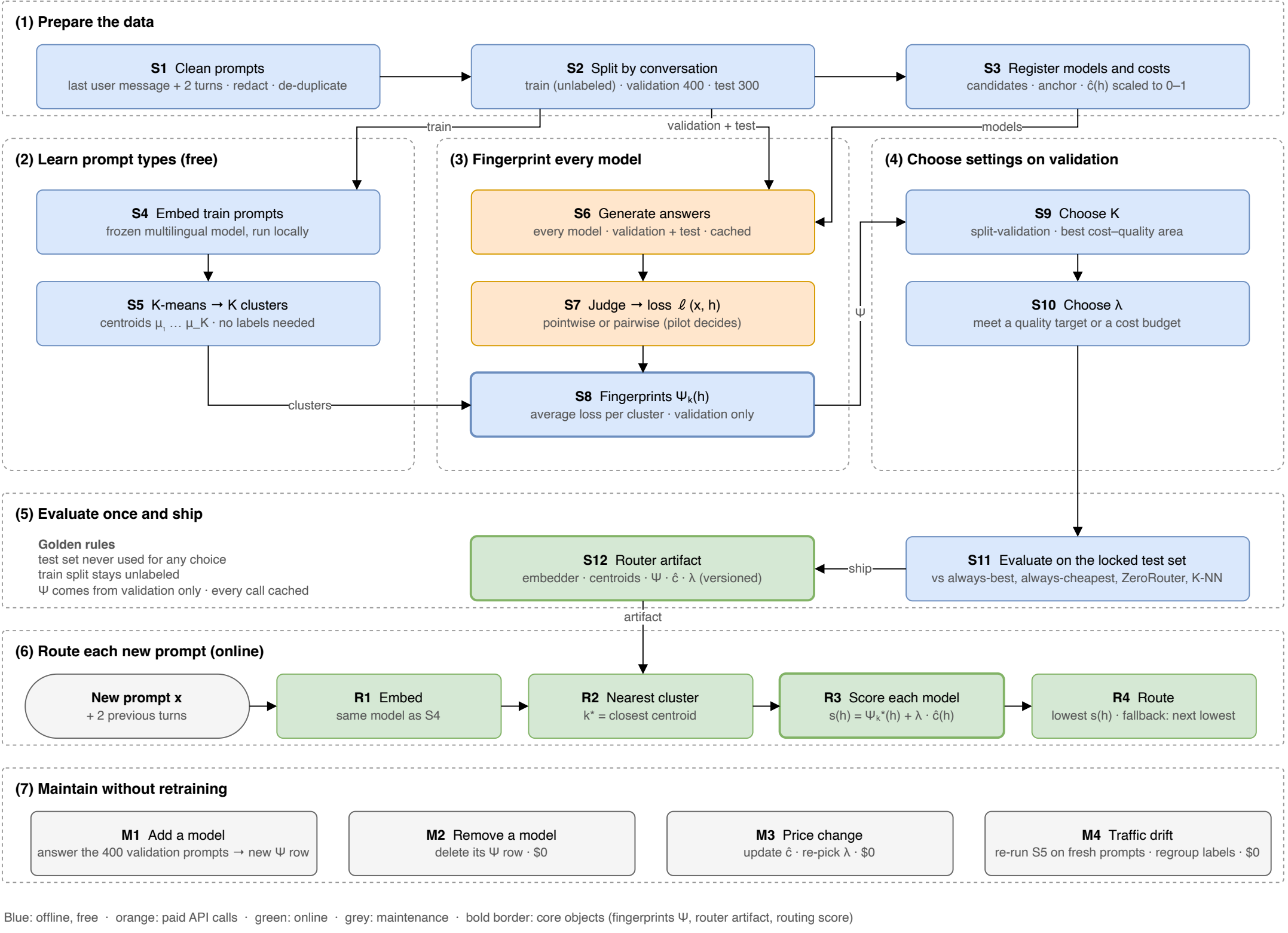

(1) Prepare the data
S1 Clean prompts
last user message + 2 turns · redact · de-duplicate
S2 Split by conversation
train (unlabeled) · validation 400 · test 300
S3 Register models and costs
candidates · anchor · ĉ(h) scaled to 0–1
train
validation + test
models
(2) Learn prompt types (free)
S4 Embed train prompts
frozen multilingual model, run locally
S5 K-means → K clusters
centroids μ_1 … μ_K · no labels needed
clusters
(3) Fingerprint every model
S6 Generate answers
every model · validation + test · cached
S7 Judge → loss ℓ (x, h)
pointwise or pairwise (pilot decides)
S8 Fingerprints Ψ_k(h)
average loss per cluster · validation only
Ψ
(4) Choose settings on validation
S9 Choose K
split-validation · best cost–quality area
S10 Choose λ
meet a quality target or a cost budget
(5) Evaluate once and ship
Golden rules
test set never used for any choice
train split stays unlabeled
Ψ comes from validation only · every call cached
S12 Router artifact
embedder · centroids · Ψ · ĉ · λ (versioned)
ship
S11 Evaluate on the locked test set
vs always-best, always-cheapest, ZeroRouter, K-NN
artifact
(6) Route each new prompt (online)
New prompt x
+ 2 previous turns
R1 Embed
same model as S4
R2 Nearest cluster
k* = closest centroid
R3 Score each model
s(h) = Ψ_k*(h) + λ · ĉ(h)
R4 Route
lowest s(h) · fallback: next lowest
(7) Maintain without retraining
M1 Add a model
answer the 400 validation prompts → new Ψ row
M2 Remove a model
delete its Ψ row · $0
M3 Price change
update ĉ · re-pick λ · $0
M4 Traffic drift
re-run S5 on fresh prompts · regroup labels · $0
Blue: offline, free · orange: paid API calls · green: online · grey: maintenance · bold border: core objects (fingerprints Ψ, router artifact, routing score)